Q. No. 1: a) Consider the following **exhaustive search** algorithm for DFS traversal of a graph.

SO : 1	Marks : 10

ALGORITHM $DFS(G)$

$count \leftarrow 0$

for each vertex v in V **do**

  if v is marked with 0

    $dfs(v)$

ALGORITHM $dfs(v)$

$count \leftarrow count + 1$;

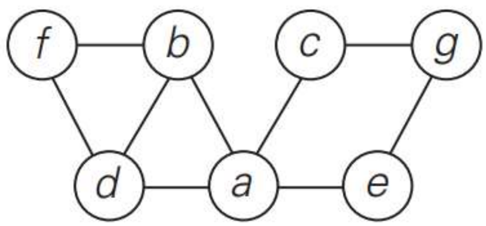
mark v with $count$

for each vertex w in V adjacent to v **do**

  if w is marked with 0

    $dfs(w)$

Applying this algorithm on the following graph, find the **DFS traversal** of the following graph (**visit vertices alphabetically**), Show all the recursive calls in the order they are invoked.



Solution: (04 pts)

S.No.	Recursive Call
1	
2	
3	
4	
5	
6	
7	

Pop-Off Sequence: (03 pts)

Traversal Sequence: (03 pts)

b) Using **exhaustive search** approach, solve the knapsack problem by **including all possibilities**, for the following set of items along with their weights and values. Capacity of knapsack is 5 Kg.

SO : 1	Marks : 5

Items	Weights (kg)	Values ($)
1	2	12
2	1	10
3	3	20
4	2	15

Solution:

Subset	Total Weight	Total Value	Feasible / Infeasible

Subset	Total Weight	Total Value	Feasible / Infeasible

Subset	Total Weight	Total Value	Feasible / Infeasible

Subset	Total Weight	Total Value	Feasible / Infeasible

Answer: Subset: Weight: Value:

Q. No. 2: Apply the following **decrease and conquer** algorithm to find all lexicographic permutations of G, M, D. You are required to fill in the following table by writing the permutations found along with the values of 'i' and 'j' at each iteration.

SO : 1	Marks : 10

ALGORITHM *LexicographicPermute*(n)
initialize the first permutation with 12 . . . n
while last permutation has two consecutive elements in increasing order **do**
  let i be its largest index such that $a_i < a_{i+1}$
  find the largest index j such that $a_i < a_j$
  swap a_i with a_j
  reverse the order of the elements from a_{i+1} to a_n inclusive
  add the new permutation to the list

Solution:

Permutation No.	0	1	2	i	j
1					
2					
3					
4					
5					
6					

Q. No. 3: (a) Apply **Decrease and Conquer** Russian peasant algorithm using to find the multiplication of 65 and 50

Solution:

SO : 1	Marks : 10

n	m	
65	50	
Final Answer:		

(b) Apply **Decrease and Conquer** interpolation search algorithm to find the value **v = 8** in the following array A:

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
5	6	8	15	20	25	30	35	40

Note: Formula to calculate 'x', is as follows:

$$x = l + \left\lfloor \frac{(v - A[l])(r - l)}{A[r] - A[l]} \right\rfloor$$

Solution:

SO : 1	Marks : 10

ℓ	r	v	A[ℓ]	A[r]	X	A[X]

SO : 1	Marks : 5

Q. No. 4: (a) Analyze the given algorithm using RAM model and find its running time.

```
ALGORITHM  BruteForcePolynomialEvaluation(P[0..n], x)
    //Computes the value of polynomial P at a given point x
    //by the "highest to lowest term" brute-force algorithm
    //Input: An array P[0..n] of the coefficients of a polynomial of degree n,
    //          stored from the lowest to the highest and a number x
    //Output: The value of the polynomial at the point x
    p ← 0.0
    for i ← n downto 0 do
        power ← 1
        for j ← 1 to i do
            power ← power * x
        p ← p + P[i] * power
    return p
```

Solution:

(b) Using **Master theorem**, solve the following recurrence relation. Write the values of all variables involved in the calculations.

SO : 1	Marks : 5

$$T(n) = 8T(n/2) + n$$

Solution:

Q. No. 5: Construct 2-3 tree by inserting the following keys in it:

SO : 1	Marks : 10

5 2 7 3 1 10 9 8

Note: Draw all intermediate trees before constructing final tree

Solution:

Q. No. 6 (a) Use Horner's rule to evaluate the polynomial $P(x) = x^4 - 3x^3 + 5x^2 - x + 2$ at $x = 2$

SO : 1	Marks : 5

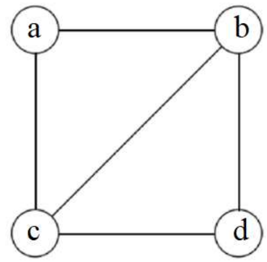
Solution:

At $x = 2$

Coefficient					
Expression					
Value					

(b) Find the number of paths indicated between the vertices of the given graph using the transform and conquer (problem reduction) technique.

SO : 1	Marks : 10



Solution:

Path of length 1:

$A^1 =$

	a	b	c	d
a				
b				
c				
d				

Path of length 2:

$A^2 =$

	a	b	c	d
a				
b				
c				
d				

Path of length 3:

$A^3 =$

	a	b	c	d
a				
b				
c				
d				

Q. No. 7(a) Design a **recursive algorithm** to return the number of leaves of any binary search tree.

SO : 2	Marks : 10

Solution:

(b) Design a **recursive algorithm** for the post order traversal of a binary search tree. What is the running time of searching any key in this tree?

SO : 2	Marks : 10

Solution:

Rough Work